

2024 Paper 2 Questions by Topic

8 (a) State the name of the class (group) of fundamental particles that contains a neutrino.

lepton(s)	B1	Allow slight misspellings e.g. 'leptones' [1]
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(b) A hadron P has a charge of $+1e$, where e is the elementary charge. The hadron P is composed of a down antiquark and only one other quark.

(i) Identifv a possible flavour for this other quark.

up or top or charm	B1	Allow 'u', 't', 'c' Ignore a <u>single</u> antidown presented alongside u, t or c ... Ignore any charge assigned to the quark [1]
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(ii) State what type of hadron is P.

.. **Meson(s)** [1]

(c) Nucleus Q undergoes radioactive decay to form nucleus R, emitting an antineutrino and another particle X, as shown in the decay equation.

$$Q \rightarrow R + X + \bar{\nu}$$

(i) State what particle is represented by X.

β^- (particle) OR electron [1]

(ii) Compare the nucleon numbers of Q and R.

..... **equal** [1]

(iii) Compare tne charges of Q and R.

(the charge of) R is greater (than Q) [1]

[Total: 6]

4 A nucleus P undergoes α -decay to form nucleus Q.

(a) Complete the equation for this decay.



(b) (i) State the principle of conservation of momentum.

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for an isolated system / no (resultant) external force

[2]

(ii) Before the decay, nucleus P has a speed of $3.2 \times 10^5 \text{ ms}^{-1}$. After the decay, nucleus Q is stationary.

Calculate the speed of the alpha particle after the decay.

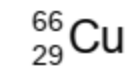
$p_{\alpha} = p_P - p_Q$	
$4(u)v = 215(u) \times 3.2 \times 10^5 (-0)$	C1
$v = 215(u) \times 3.2 \times 10^5 / 4(u)$	
$v = 1.7 \times 10^7 \text{ m s}^{-1}$	A1

speed = ms^{-1} [2]

[Total: 6]

7 Nuclei of an isotope of copper (Cu) each have 29 protons and 37 neutrons. This isotope is a β^- emitter.

(a) State the nuclide notation in the form ${}^A_Z\text{X}$ for this nucleus of copper.



[1]

(b) The energy spectrum of the β^- radiation emitted by a sample of this isotope is shown in Fig. 7.1.

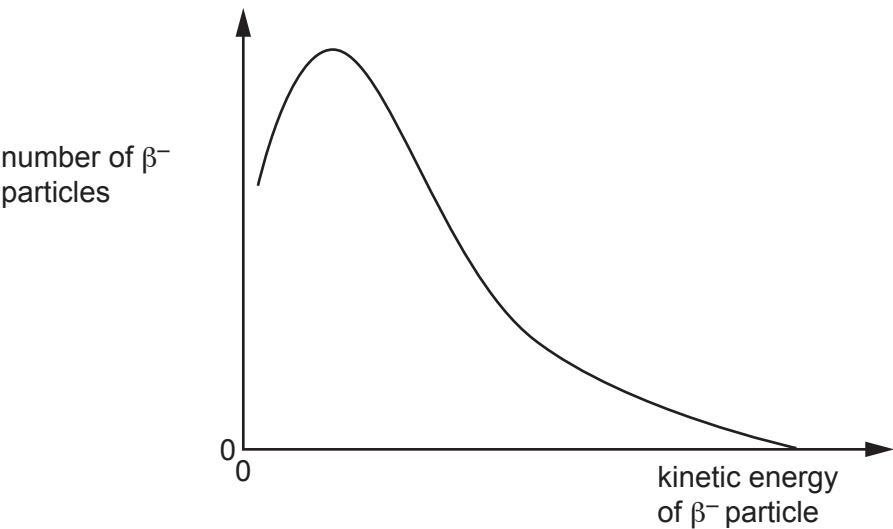


Fig. 7.1

(i) Use Fig. 7.1 to explain why other particles apart from the β^- particles must be emitted during this decay.

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the energy of the decay is fixed / constant	B1	
the energies of the beta particles have a (continuous) range of values / varies / not constant	B1	
another particle / an (anti)neutrino must possess the extra / remaining energy (difference between energy of the decay and the β kinetic energy)	B1	

..... [3]

(ii) State the name of the other particle emitted during the decay of this isotope.

(electron) antineutrino	B1	[1]
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(iii) The copper isotope decays to an isotope of zinc (Zn).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

${}^{66}_{29}\text{Cu} \rightarrow {}^{66}_{30}\text{Zn} + {}^0_{-1}\beta + {}^0_0\overline{\nu}_e$ values for Cu and Zn correct with no other extra particles on either side of the equation	B1
second term correct (${}^0_{-1}\beta$)	B1
third term correct (${}^0_0\overline{\nu}_e$)	B1

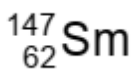
[3]

[Total: 8]

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6 Nuclei of an isotope of samarium (Sm) each contain 62 protons and 85 neutrons.

(a) State the nuclide notation in the form A_ZX for this isotope of samarium.



[1]

(b) This isotope of samarium is radioactive and decays by emitting particles. Gamma-radiation is **not** emitted. The energy spectrum of the emitted particles is shown in Fig. 6.1.

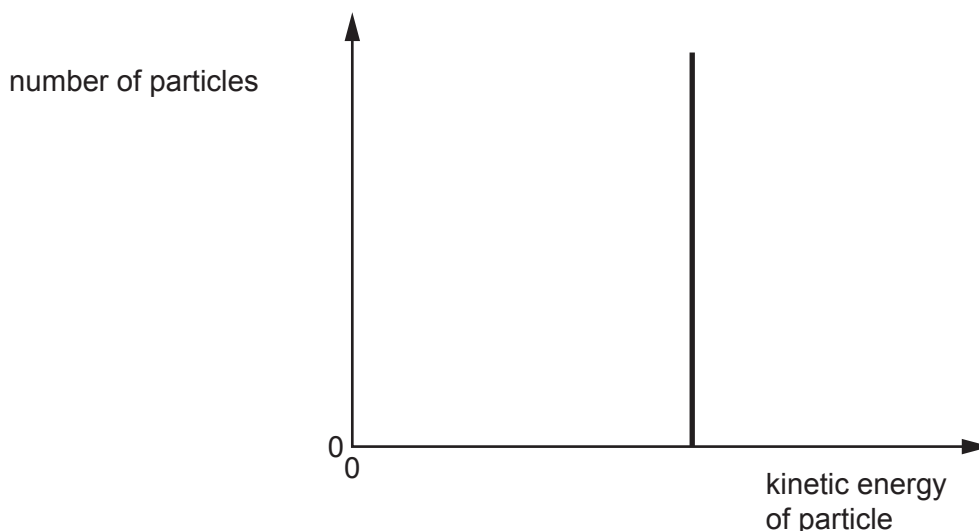


Fig. 6.1

(i) Explain how Fig. 6.1 shows that this isotope of samarium emits α -particles and does **not** emit β -particles.

.....
the (kinetic) energy of the particles is discrete / has only one value (so must be alpha)

.....
and beta particles have a (continuous) range of (kinetic) energies (so can't be beta)

[2]

(ii) This isotope of samarium decays to an isotope of neodymium (Nd).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

${}^{147}_{62}\text{Sm} \rightarrow {}^{143}_{60}\text{Nd} + {}^4_2\alpha$	A1
values for Sm and Nd correct with no other extra particles on either side of the equation	
${}^4_2\alpha$ correct	A1

[2]

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(c) A baryon is composed of three quarks which all have different flavours. The baryon has a charge of 0.

Two of the quarks in the baryon are an up quark and a bottom quark.

(i) Determine, in terms of the elementary charge e , the charge on the third quark in the baryon.

up quark charge is $+(2/3)(e)$ or bottom quark charge is $-(1/3)(e)$	C1
$0 = +(2/3)(e) - (1/3)(e) + q$ (so) charge (on third quark must be) $-(1/3)(e)$	A1

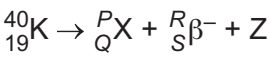
charge = e [2]

(ii) State a possible flavour for the third quark in the baryon.

down or strange	A1	[1]
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[Total: 8]

- 5 (a) Potassium-40 ($^{40}_{19}\text{K}$) undergoes β^- decay to form a nuclide of element X. Particle Z is emitted during the decay. The equation for the decay is shown.



- (i) State the values of P , Q , R and S .

$P =$

$R =$

$Q =$

$S =$

$$P = 40 \text{ and } R = 0$$

$$Q = 20 \text{ and } S = -1$$

[2]

- (ii) State the name of particle Z.

(electron) antineutrino [1]

- (iii) State the name of the class of fundamental particle to which both the β^- particle and particle Z belong.

..... leptons [1]

- (b) Determine the quark composition of an alpha-particle.

(alpha particle consists of) 2 protons and 2 neutrons	C1
quark composition of proton = up up down or quark composition of neutron = up down down	C1
quark composition (of alpha particle) = 6 up, 6 down	A1

quark composition [3]

[Total: 7]

6 (a) Compare an α -particle with a β^+ particle in terms of their masses and charges.

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.....

.....

.....

..... [3]

(b) Nucleus P undergoes α -decay to form nucleus Q. Nucleus Q then undergoes a further decay to form nucleus R. The proton and nucleon numbers of P and R are shown in Fig. 6.1.

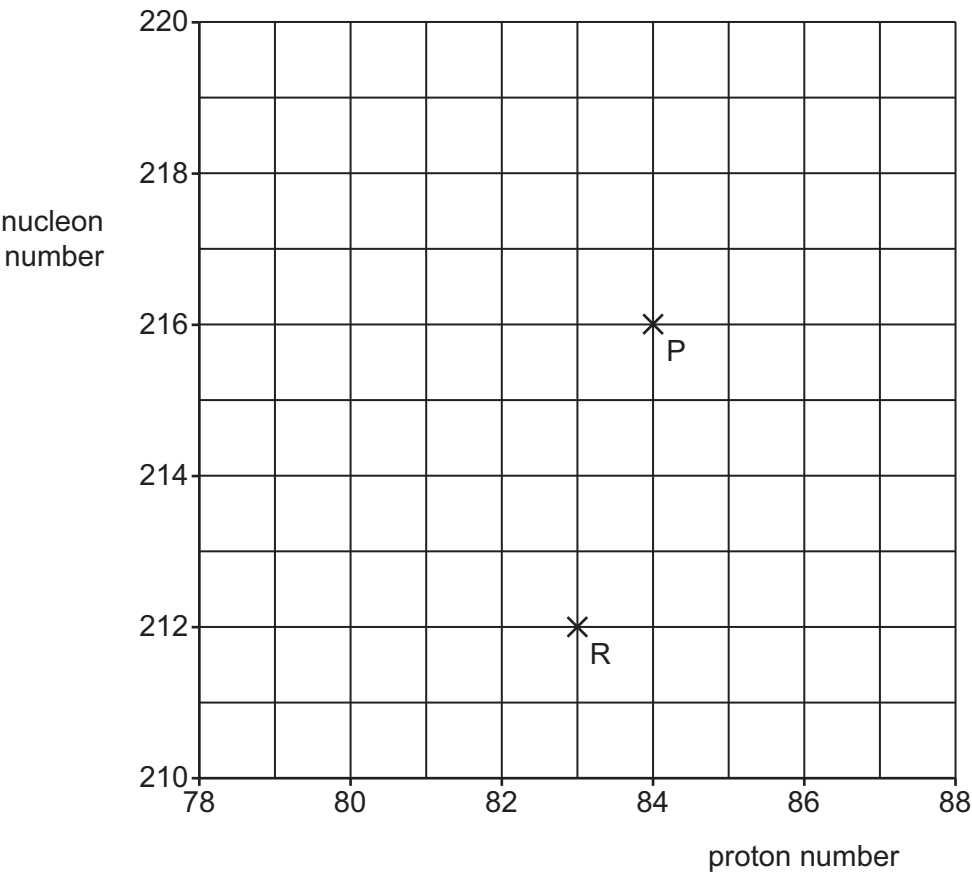


Fig. 6.1

cross labelled Q at (82, 212)	B1
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(i) On Fig. 6.1, draw a cross (×) to show the proton number and nucleon number of Q. Label your cross Q. [1]

(ii) State the names of the particles emitted as Q decays to form R.

particles emitted are: • beta-minus (particle) / electron • (electron) antineutrino	B1	
either particle named		 [2]
both particles named and no incorrect particles named	B1	

- (c) Before the α -decay, P is travelling at a constant velocity. After the decay, Q has a velocity of $1.3 \times 10^5 \text{ m s}^{-1}$ at an angle of 68° to the original path of P. The α -particle has a velocity of $150 \times 10^5 \text{ m s}^{-1}$ at an angle of θ to the original path of P, as shown in Fig. 6.2.

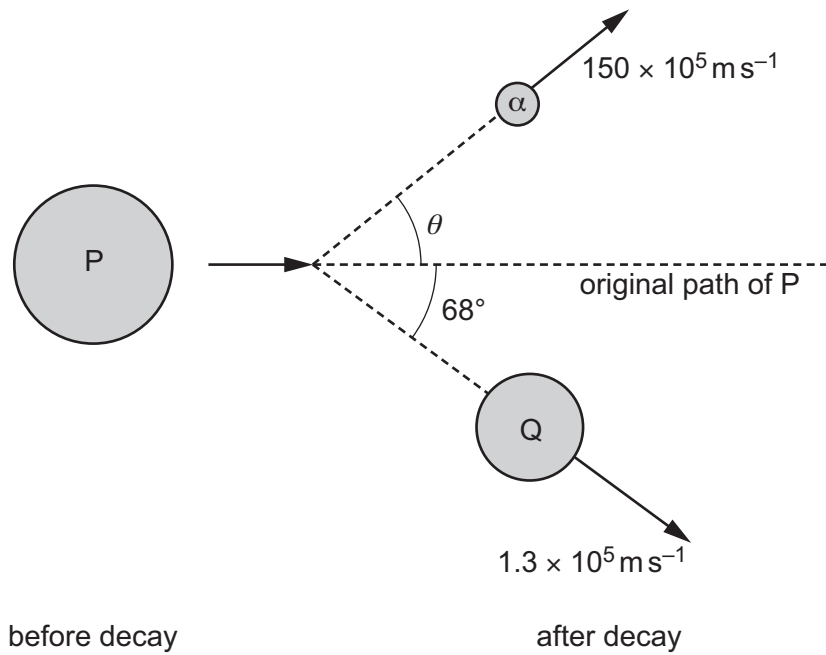


Fig. 6.2 (not to scale)

- (i) Use the principle of conservation of momentum to determine θ .

$212(u) \times 1.3 (\times 10^5) \sin 68^\circ$ or $4(u) \times 150 (\times 10^5) \sin \theta$	C1
$4(u) \times 150 (\times 10^5) \times \sin \theta = 212(u) \times 1.3 (\times 10^5) \times \sin 68^\circ$	C1
$\sin \theta = 0.426$ $\theta = 25^\circ$	A1

$\theta = \dots\dots\dots^\circ$ [3]

- (ii) Calculate the kinetic energy of the α -particle.

$E = \frac{1}{2}mv^2$	C1
$= \frac{1}{2} \times 4 \text{ u} \times (150 \times 10^5)^2$ $= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (150 \times 10^5)^2$ $= 7.5 \times 10^{-13} \text{ J}$	A1

kinetic energy = $\dots\dots\dots$ J [2]

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- 7 (a) Complete Table 7.1 to show the charges, in terms of the elementary charge e , on each of the flavours of quark and antiquark shown.

Table 7.1

flavour	charge / e		
	quark	antiquark	
up	up quark charge = (+) 2 / 3 and down quark charge = −1 / 3		B1
	strange quark charge = −1 / 3		B1
down	up antiquark charge = −2 / 3 and down antiquark charge = (+) 1 / 3 and strange antiquark charge = (+) 1 / 3		B1
strange			

[3]

- (b) (i) State the name of the class (group) of fundamental particles to which baryons and mesons belong.

..... **hadron(s)** [1]

- (ii) Compare baryons and mesons in terms of their constituent particles.

baryons composed of three quarks or baryons composed of three antiquarks	B1	
mesons composed of one quark and one antiquark	B1	

..... [2]

- (c) Describe β^+ decay in terms of the fundamental particles involved.

up quark changes to a down quark	B1
positron and (electron) neutrino (emitted)	B1

.....
..... [2]

[Total: 8]

7 A particle Q and a particle R are each composed of one quark and one antiquark.

(a) State the name of the class (group) of particles that includes Q and R.

.. meson(s) or hadron(s) [1]

(b) Q has a charge of $-1e$, where e is the elementary charge. R has a charge of 0.

Complete Table 7.1 to show a possible second quark in each of Q and R.

Table 7.1

	charge	first quark	second quark
Q	$-1e$	strange	
R	0	anti-up	

[2]

Q: anti-up or anti-charm or anti-top

[Total: 3]

R: up or charm or top